SEMICONDUCTORS & ELECTRONICS COURSE — 2025/2026

Electronic Alarm System

Using Transistors, RC Timing & Flame Detection

FINAL REPORT — MAY 2026

Project	**Project 3 — Simple Electronic Alarm Using Transistors and RC Timing**
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1. INTRODUCTION

This report presents the complete work carried out by our group . The project consists of designing and implementing a Simple Electronic Alarm System using transistors, RC timing, and flame detection.

The alarm activates when infrared radiation from a flame is detected by a photodiode which is a PN junction semiconductor device. The signal is filtered through an RC network to prevent false triggers, then applied to the base of an NPN bipolar junction transistor , which switches ON and drives an active buzzer.

2. THEORETICAL BACKGROUND

2.1 The Flame Sensor or IR Photodiode

The flame sensor is an infrared photodiode which is a PN junction device that detects IR radiation in the 760 nm to 1100 nm wavelength range emitted by flames. When IR photons strike the depletion region of the PN junction, they transfer energy to valence electrons, pushing them into the conduction band. This generates a photocurrent proportional to the IR intensity, causing the sensor output voltage to vary with the presence or absence of flame.

Key principle: Under forward bias, the depletion region of the PN junction narrows and current flows. The built-in potential $V_b \approx 0.7V$ for silicon must be overcome to allow conduction ,this threshold directly determines when the transistor switches ON.

2.2 NPN Transistor : Its Switching Behavior

The NPN bipolar junction transistor 2N2222 consists of two PN junctions: the Base-Emitter (BE) junction and the Base-Collector (BC) junction. In this circuit it operates as an electronic switch with two states:

State	Condition	Result
OFF (Cut-off)	$V_{BE} < 0.7V$: no flame detected	No collector current ⟶ Buzzer silent
ON (Saturation)	$V_{BE} > 0.7V$: flame detected	Collector current flows ⟶ Buzzer ON

2.3 RC Timing Network

The RC network ($R_2 = 10k\Omega$, $C_1 = 10\mu F$) introduces a time delay $\tau = R \times C = 10{,}000 \times 0.0001 =$ **1 second**. When a flame is detected, the capacitor slowly charges through R_2. The alarm only triggers when the voltage across C_1 exceeds $V_{BE} = 0.7V$. When the flame disappears, C_1 discharges slowly and the alarm stays ON for approximately 1 second before turning OFF. This prevents false triggers from brief flame flickers.

3. CIRCUIT DESIGN

3.1 System Architecture

The alarm system operates in three sequential stages:

TRIGGER
IR Photodiode
(Flame Sensor)

⟶

SWITCH
NPN Transistor
+ RC Delay (1s)

⟶

OUTPUT
Active Buzzer
5V DC

3.2 Component List

Component	Value	Qty	Role in Circuit
IR Photodiode	Flame sensor	1	Detects IR radiation — PN junction device
NPN Transistor	2N2222	1	Electronic switch — two PN junctions
Base Resistor R1	1 kΩ	1	Limits base current, protects transistor
RC Resistor R2	10 kΩ	1	Controls RC time constant τ = 1s
RC Capacitor C1	100 µF	1	Timing delay : prevents false triggers

Active Buzzer	5V DC	1	Audio alarm output
Arduino UNO R3	ATmega328P	1	5V power supply + signal processing
Breadboard	830 tie-points	1	Prototype assembly platform

3.3 Circuit Schematic

The Arduino UNO provides a regulated 5V supply. The IR photodiode anode connects to 5V and its cathode connects directly to A0 on the Arduino. The analog signal is fed to the RC network (R2 + C1) which introduces a 1-second delay. The filtered signal is then applied to the transistor base through R1. When V^{be} exceeds 0.7V, the transistor saturates and current flows through the buzzer, triggering the alarm.

CIRCUIT SCHEMATIC (simplified)

```
+5V
  |
  +--[Photodiode Anode (+)]
     [Photodiode Cathode (-)]--A0
                              |
                            [R2:10k]--[C1:100uF]--[R1:10k]--> Base(B)
                                                              NPN 2N2222
+5V--[BUZZER+]--Collector(C)          Emitter(E)--GND
```

3.4 Arduino Code

The following code reads the photodiode signal and controls the buzzer:

```
int sensorPin = A0;
int buzzerPin = 8;
int sensorValue = 0;

void setup() {
  pinMode(buzzerPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  sensorValue = analogRead(sensorPin);
  Serial.print("Sensor: ");
  Serial.println(sensorValue);

  if (sensorValue > 200) {
    tone(buzzerPin, 1000);
    Serial.println(">>> FLAME DETECTED - ALARM ON");
  } else {
    noTone(buzzerPin);
    Serial.println("No flame");
  }
  delay(200);
}
```

4. SIMULATION RESULTS

The circuit was simulated using Tinkercad Circuits. The photodiode simulates the IR flame sensor behavior. The light intensity slider controls the sensor output, simulating the presence or absence of a flame.

4.1 No Flame — Alarm OFF

When light intensity is reduced (no flame), the sensor value drops to 140 — below the threshold of 200. The Serial Monitor confirms: Sensor: 140 : No flame. The transistor returns to cut-off mode and the buzzer is silent.

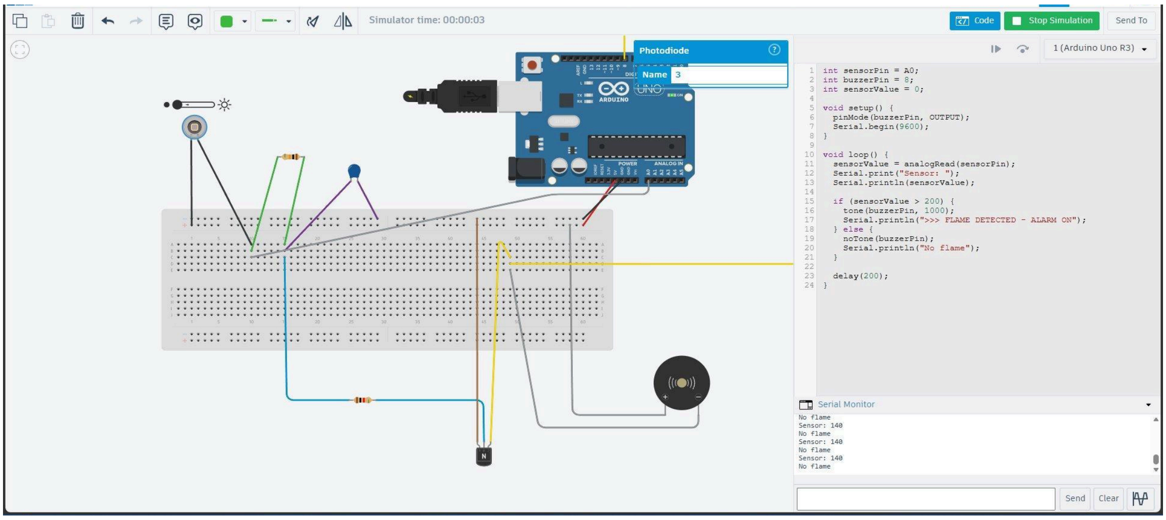

Figure 2 — Tinkercad simulation: No flame. Serial Monitor shows Sensor: 140 and No flame.

4.2 Flame Detected — ALARM ON

When light intensity is increased (simulating flame), the sensor value rises above the threshold of 200. The Serial Monitor confirms: Sensor: 373 — FLAME DETECTED — ALARM ON. The buzzer activates and the transistor is in saturation mode.

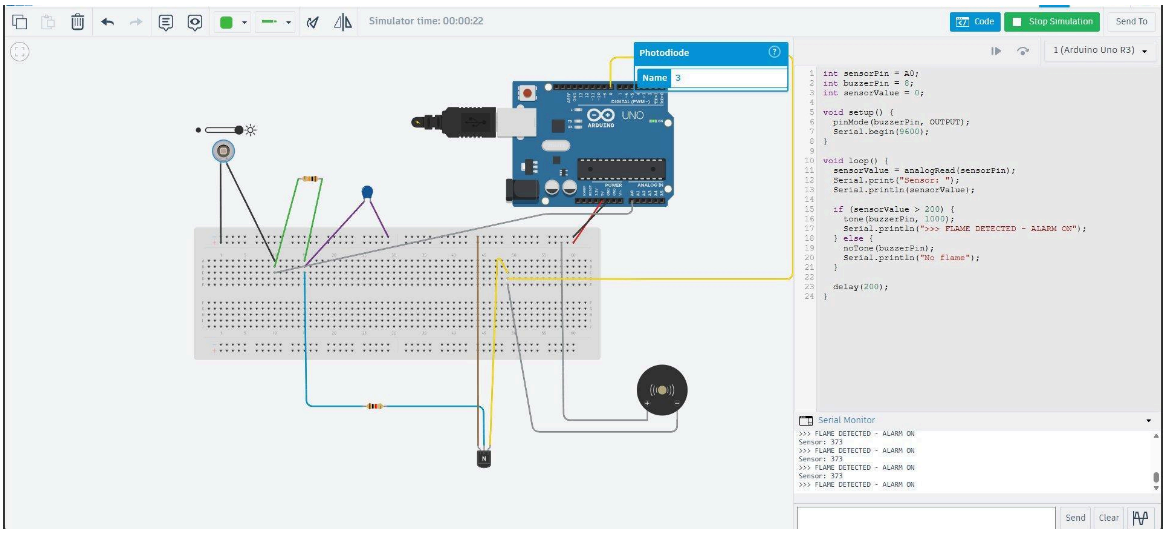

Figure 2 — Tinkercad simulation: Flame detected. Serial Monitor shows Sensor: 373 and ALARM ON.

4.3 Results Summary

Condition	Sensor Value	Transistor State	Buzzer
Flame detected	373 (> 200)	**Saturation (ON)**	**Sounding** ✅
No flame	140 (< 200)	Cut-off (OFF)	Silent ❌

5. CONCLUSION

This project successfully demonstrated the design and simulation of a simple electronic alarm system based on semiconductor devices. The IR photodiode — a PN junction device — detects flame by converting infrared radiation into an electrical signal. The RC timing network filters the signal with a 1-second delay ($\tau = R \times C = 1s$), preventing false triggers. The NPN transistor 2N2222 acts as a switch, transitioning from cut-off to saturation when the base-emitter voltage exceeds the built-in potential V^b = 0.7V of the silicon PN junction — directly confirming the theory studied in Chapters 1 and 2.

The Tinkercad simulation confirmed the theoretical calculations: sensor value 373 triggers the alarm while sensor value 140 keeps it silent, with threshold set at 200. The circuit will be physically implemented on a breadboard using the Arduino UNO R3 Starter Kit, replacing the simulation photodiode with a real 2-pin IR flame sensor.